FIGURE 2

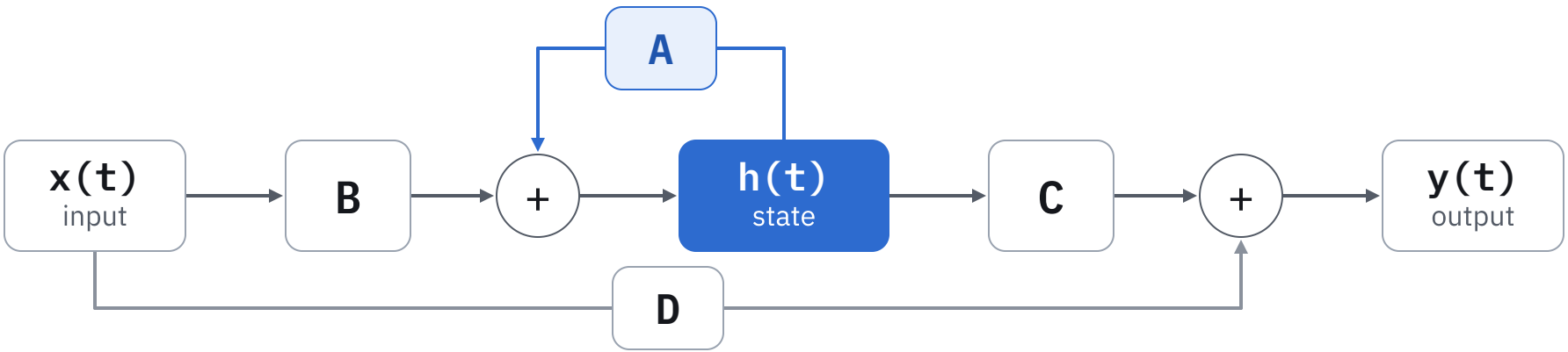
A state space model: two equations, four matrices

$$h'(t) = A \cdot h(t) + B \cdot x(t)$$

state equation: how the memory changes

$$y(t) = C \cdot h(t) + D \cdot x(t)$$

output equation: how the memory is read



A carries the past forward on its own

B writes the new input into the state

C reads the part of the state that matters now

D a shortcut from input straight to output